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NDIC - DDR XML - Common traffic information

Release 3.2.5rev1

Národní Dopravní Informační Centrum

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CONTENTS

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The specification defines format of data distribution interface (DDR) of NDIC used for publishing of common traffic information (accidents etc.) and traffic statuses.

CHAPTER

ONE

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format and other schemas (RELAX NG)
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 Changelog

version 3.2.5

- latest revision of the format, additionally added information about directionality error at ALERT-C location.

1.2 General terms

Following terms are used in this document:

format
description of data structure for transferring some kind of information

protocol
description of data transfer process and rules

NDIC
national traffic information centre

DDR
data distribution interface, a proprietary format used by NDIC

(W3C) XML schema
language, which allows definition of rules for a XML document structure.

RELAX NG schema
human readable language, which allows definition of rules for a XML document structure.

XSD
XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri
uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CHAPTER

TWO

CONCEPTS

This section highlights main concepts, used by this format. For complete description refer to schemas with definitions in comments.

In this section are provided some snippets of real XML documents. For more examples, see *Appendix A: Message examples*

2.1 XML Document Sections

The XML Document is built using following elements and sections:

- DOC (document root)
- INF (metadata)
- MJD (container for messages)
- MSG (message)

2.1.1 Document root (*DOC*)

The *DOC* element represents single publication (one instance of information message).

Attribute *id* provides unique identity to given publication as a whole.

```xml
<?xml version="1.0" encoding="utf-8"?>
<DOC DataSet="extended" country="CZ" id="7e78a1b7-86c1-41be-b980-442d447f7b16" version="3.0">
  <INF receiver="JB007" sender="JSDI_NDIC" transmission="HTTP">
    <DAT>
      <EVTT language="CZ" version="2.01"/>
      <LOCT country="2" number="25" version="5.0"/>
      <SNET country="CZ" type="GN" version="14.12"/>
      <UIRADR structure="4.2" version="972"/>
    </DAT>
  </INF>
  <MJD count="1">
```

2.1.2 Metadata (*INF*)

INF element provides meta-information about whole publication:

- reference to ALERT-C event table version
- reference to ALERT-C location table version used for describing locations
- reference to GDF 4.0 map data set publisher and version
- reference to state organization units and areas enumeration UIR-ADR

```xml
<INF receiver="JB007" sender="JSDI_NDIC" transmission="HTTP">
  <DAT>
    <EVTT language="CZ" version="2.01"/>
    <LOCT country="2" number="25" version="5.0"/>
    <SNET country="CZ" type="GN" version="14.12"/>
    <UIRADR structure="4.2" version="972"/>
  </DAT>
</INF>
<MJD count="1">
```

2.1.3 Message container (*MJD*)

Single traffic messages are put into one parent element *MJD*. It can contain one or more messages.

2.1.4 Traffic Message (*MSG*)

Particular message is represented by a *MSG* element.

The message itself has following sections:

- validity *MTIME*
- textual description of whole event *MTXT*
- event description *MEVT* (*what* has happened)
- location reference *MLOC* (*where*) to linear location
- reference to a region in Czech Republic *MDST*

```xml
  <MSG id="{ff808181-5024-3057-0150-44cdefe07f8a}" planned="false" type="TI" version="6">
    <MTIME format="YYYY-MM-DDThh:mm:ssTZD">
      <TGEN>2015-10-08T03:13:12+02:00</TGEN>
      <TSTA>2015-10-08T02:10:00+02:00</TSTA>
      <TSTO>2015-10-08T03:55:00+02:00</TSTO>
    </MTIME>
    <MTXT language="CZ">Od 8.10.2015 02:10 do 03:55; na silnici 13 u obce
    Frýdlant okres Liberec; havarované vozidlo; probíhá vyšetřování
    nehody; havárie OA, se zraněním, složky IZS jsou na místě.</MTXT>
    <MEVT>
      <TMCE directionalityvalue="1" diversion="false" timescalevalue="(D)" urgencyvalue="N">
        <EVI eventcode="211" eventorder="1" updateclass="4">
          <TXUCL language="CZ">Dopravní události</TXUCL>
          <TXEVC language="CZ">havarované vozidlo</TXEVC>
        </EVI>
        <EVI eventcode="344" eventorder="2" updateclass="3">
```

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```
          <TXUCL language="CZ">Nehody</TXUCL>
          <TXEVC language="CZ">probíhá vyšetřování nehody</TXEVC>
        </EVI>
        <TXTMCE language="CZ">havarované vozidlo probíhá vyšetřování
        nehody</TXTMCE>
      </TMCE>
      <OTXT>havarované vozidlo; probíhá vyšetřování nehody; havárie OA,
      se zraněním, složky IZS jsou na místě.</OTXT>
    </MEVT>
    <MLOC>
      <TXPL>na silnici 13 u obce Frýdlant okres Liberec</TXPL>
      <TMCL direction="+" extent="1" primarycode="16366" roadid="332"/>
      <SNTL coordsystem="S-JTSK" count="1">
        <COORD x="-687866" y="-962713"/>
        <STEL el_code="2303555"/>
      </SNTL>
    </MLOC>
    <MDST>
      <DEST CountryName="Česká republika" RegionCode="78" RegionName="Liberecký" TownCode="564028" TownName="Frýdlant" TownShip="Liberec" TownShipCode="3505">
        <ROAD RoadClass="1" RoadNumber="13"/>
      </DEST>
    </MDST>
  </MSG>
```

Message identity and version

Message itself has unique identifier *id* and version. Version shall start from *0* and be incremented by each update.

Validity of the message in time is specified by simple time range from start time to end time.

```
  <MSG id="{ff808181-5024-3057-0150-44cdefe07f8a}" planned="false" type="TI" version="6">
    <MTIME format="YYYY-MM-DDThh:mm:ssTZD">
      <TGEN>2015-10-08T03:13:12+02:00</TGEN>
      <TSTA>2015-10-08T02:10:00+02:00</TSTA>
      <TSTO>2015-10-08T03:55:00+02:00</TSTO>
    </MTIME>
```

Event description

DDR is based in ALERT-C event catalogue and follows all the ALERT-C rules (multiple event codes, quantifiers…

Apart from ALERT-C, free textual description can be used on many places.

```
    <MEVT>
      <TMCE directionalityvalue="1" diversion="false" timescalevalue="(D)" urgencyvalue="N">
        <EVI eventcode="211" eventorder="1" updateclass="4">
          <TXUCL language="CZ">Dopravní události</TXUCL>
          <TXEVC language="CZ">havarované vozidlo</TXEVC>
        </EVI>
        <EVI eventcode="344" eventorder="2" updateclass="3">
          <TXUCL language="CZ">Nehody</TXUCL>
          <TXEVC language="CZ">probíhá vyšetřování nehody</TXEVC>
```

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```
          </EVI>
          <TXTMCE language="CZ">havarované vozidlo probíhá vyšetřování
          nehody</TXTMCE>
        </TMCE>
        <OTXT>havarované vozidlo; probíhá vyšetřování nehody; havárie OA,
        se zraněním, složky IZS jsou na místě.</OTXT>
      </MEVT>
```

Location reference

DDR is using only point and linear locations on road network.

There are following location referencing methods used:

- Free textual description
- ALERT-C location codes (incl. direction and extent).
- Reference to sequence of road segments as defined in GDF 4.0 based map data set.
- Coordinate of head of the event
- Reference to region in Czech Republic using enumeration UIR-ADR.
- Road reference (where possible using mileage)

```
      <MLOC>
        <TXPL>na silnici 13 u obce Frýdlant okres Liberec</TXPL>
        <TMCL direction="+" extent="1" primarycode="16366" roadid="332"/>
        <SNTL coordsystem="S-JTSK" count="1">
          <COORD x="-687866" y="-962713"/>
          <STEL el_code="2303555"/>
        </SNTL>
      </MLOC>
      <MDST>
        <DEST CountryName="Česká republika" RegionCode="78" RegionName="Liberecký" TownCode="564028" TownName="Frýdlant" TownShip="Liberec" TownShipCode="3505">
          <ROAD RoadClass="1" RoadNumber="13"/>
        </DEST>
      </MDST>
```

CHAPTER

THREE

LOCATION REFERENCING

Following is a selection of most used location referencing methods.

3.1 Location description by means of ALERT-C

The first method describes position by ALERT-C codes, by element *TMCL*. The use of this method depends on the presence of the ALERT-C location at a given location.

The *TMCL* method uses the primary location (attribute *primarycode*), extent (attribute *extent*) and direction (attribute`direction`) element to describe the event. Used location table is described at the level of document in element *INF*, so all messages *MSG* in the document must use same location table.

The primary location is, by the ALERT-C principle, the place where the accident occurred or where the driver leaves the problem area. The secondary location is the end of the traffic queue or the place where the driver first enters the problem area.

```
        <TMCL direction="+" extent="1" primarycode="16366" roadid="332"/>
```

Important: The *direction* attribute is incorrectly reported. The value "-" is used instead of "+" and vice versa. We do not fix this error because we assume it is already historically incorporated in the systems of the subscribers. When implementing ALERT-C localization, please reverse the direction given in the file.

3.2 Location description using the Global Network

The second method uses for location description identifiers of affected sections of the road network, specified by Global Network (GN) database, and event start and (end) coordinates (in the S-JTSK or WGS-84 reference system).

The *COORD* element describes the location of one event on the network. It is the location of the beginning of the event in the sense of traffic direction, in the specified coordinate system. For linear locations it is the tail of a traffic event, the place of first arrival or the secondary TMC location.

As an undocumented option the "end" of event can be provided in an element *COORDEND*. This element is a end of event in the sense of traffic direction, in the specified coordinate system. For linear locations it is the head of a traffic event, the place where driver leaves a problem area or the primary TMC location.

Elements *STEL* describes specific segments of the GN database by their identifier. (These elements may not have to be provided in a topological sequential order).

This is native means of location referencing by NDIC.

```
        <SNTL coordsystem="S-JTSK" count="5">
          <COORD x="-599220" y="-1163113"/>
          <STEL el_code="725704"/>
```

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```
          <STEL el_code="725706"/>
          <STEL el_code="638420"/>
          <STEL el_code="638412"/>
          <STEL el_code="638377"/>
        </SNTL>
```

3.3 Location description by enumeration UIR-ADR and road name

The third method uses the "address" of the street or road (name, area, number, etc.).

The *MDST/DEST* element describes the destination area by the administrative area affected and specific roads or streets affected. In elements *ROAD* or *STREET*.

```
      <MDST>
        <DEST CountryName="Česká republika" RegionCode="116" RegionName="Jihomoravský" TownCode="582786" TownDistrictCode="550973" TownDistrictName="Brno-střed" TownName="Brno" TownShip="Brnoměsto" TownShipCode="3702">
          <STRE StreetCode="22063" StreetName="Cejl"/>
          <STRE StreetCode="23035" StreetName="Dvorského"/>
          <STRE StreetCode="28339" StreetName="Malinovského náměstí"/>
          <STRE StreetCode="28916" StreetName="Nové sady"/>
          <STRE StreetCode="30007" StreetName="Polní"/>
          <STRE StreetCode="30031" StreetName="Ponávka"/>
          <STRE StreetCode="30481" StreetName="Příkop"/>
          <STRE StreetCode="32131" StreetName="Renneská třída"/>
          <STRE StreetCode="33081" StreetName="Nádražní"/>
          <STRE StreetCode="33529" StreetName="Benešova"/>
          <STRE StreetCode="33537" StreetName="Milady Horákové"/>
          <STRE StreetCode="34584" StreetName="Vídeňská"/>
          <STRE StreetCode="34932" StreetName="Vsetínská"/>
          <STRE StreetCode="36871" StreetName="Strážní"/>
        </DEST>
        <DEST CountryName="Česká republika" RegionCode="116" RegionName="Jihomoravský" TownCode="582786" TownDistrictCode="551031" TownDistrictName="Brno-sever" TownName="Brno" TownShip="Brnoměsto" TownShipCode="3702">
          <STRE StreetCode="27791" StreetName="Merhautova"/>
          <STRE StreetCode="33537" StreetName="Milady Horákové"/>
        </DEST>
      </MDST>
```

CHAPTER

FOUR

APPENDIX A: MESSAGE EXAMPLES

4.1 accident.xml - a vehicle accident

Publication contains exactly one message about a vehicle accident.


```xml
<?xml version="1.0" encoding="utf-8"?>
<DOC DataSet="extended" country="CZ" id="7e78a1b7-86c1-41be-b980-442d447f7b16" version="3.0">
  <INF receiver="JB007" sender="JSDI_NDIC" transmission="HTTP">
    <DAT>
      <EVTT language="CZ" version="2.01"/>
      <LOCT country="2" number="25" version="5.0"/>
      <SNET country="CZ" type="GN" version="14.12"/>
      <UIRADR structure="4.2" version="972"/>
    </DAT>
  </INF>
  <MJD count="1">
    <MSG id="{ff808181-5024-3057-0150-44cdefe07f8a}" planned="false" type="TI" version="6">
      <MTIME format="YYYY-MM-DDThh:mm:ssTZD">
        <TGEN>2015-10-08T03:13:12+02:00</TGEN>
        <TSTA>2015-10-08T02:10:00+02:00</TSTA>
        <TSTO>2015-10-08T03:55:00+02:00</TSTO>
      </MTIME>
      <MTXT language="CZ">Od 8.10.2015 02:10 do 03:55; na silnici 13 u obce
      Frýdlant okres Liberec; havarované vozidlo; probíhá vyšetřování
      nehody; havárie OA, se zraněním, složky IZS jsou na místě.</MTXT>
      <MEVT>
        <TMCE directionalityvalue="1" diversion="false" timescalevalue="(D)" urgencyvalue="N">
          <EVI eventcode="211" eventorder="1" updateclass="4">
            <TXUCL language="CZ">Dopravní události</TXUCL>
            <TXEVC language="CZ">havarované vozidlo</TXEVC>
          </EVI>
          <EVI eventcode="344" eventorder="2" updateclass="3">
            <TXUCL language="CZ">Nehody</TXUCL>
            <TXEVC language="CZ">probíhá vyšetřování nehody</TXEVC>
          </EVI>
          <TXTMCE language="CZ">havarované vozidlo probíhá vyšetřování
          nehody</TXTMCE>
        </TMCE>
        <OTXT>havarované vozidlo; probíhá vyšetřování nehody; havárie OA,
        se zraněním, složky IZS jsou na místě.</OTXT>
      </MEVT>
      <MLOC>
```


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```
        <TXPL>na silnici 13 u obce Frýdlant okres Liberec</TXPL>
        <TMCL direction="+" extent="1" primarycode="16366" roadid="332"/>
        <SNTL coordsystem="S-JTSK" count="1">
          <COORD x="-687866" y="-962713"/>
          <STEL el_code="2303555"/>
        </SNTL>
      </MLOC>
      <MDST>
        <DEST CountryName="Česká republika" RegionCode="78" RegionName="Liberecký"␣
↪TownCode="564028" TownName="Frýdlant" TownShip="Liberec" TownShipCode="3505">
          <ROAD RoadClass="1" RoadNumber="13"/>
        </DEST>
      </MDST>
    </MSG>
  </MJD>
</DOC>
```

4.2 obstacle.xml - blocked road

Publication contains exactly one message about blocked road and traffic congestion.

```
<?xml version="1.0" encoding="utf-8"?>
<DOC DataSet="extended" country="CZ" id=" 1f1502dc-86b3-405b-bfb9-dfc6a582d315"␣
↪version="3.0">
  <INF receiver="Odběratel ID" sender="JSDI_NDIC" transmission="HTTP">
    <DAT>
      <EVTT language="CZ" version="2.01"/>
      <SNET country="CZ" type="SN" version="13.12"/>
      <UIRADR structure="4.2" version="907"/>
    </DAT>
  </INF>
  <MJD count="1">
    <MSG id="eca17d6a-5eea-48e6-b61f-f6060f6ada54" planned="false" type="TI" version=
↪"1">
      <MTIME format="YYYY-MM-DDThh:mm:ssTZD">
        <TGEN>2007-09-26T08:27:19+02:00</TGEN>
        <TSTA>2007-09-26T08:27:19+02:00</TSTA>
        <TSTO>2007-10-26T08:27:19+02:00</TSTO>
      </MTIME>
      <MTXT language="CZ">Z ulice Vídeňská - Merhautova, do ulice
      Provazníkova, neprůjezdné, překážka na vozovce, dopravní kolaps v
      úseku 1 km, mimořádná událost, očekávejte zdržení, po zbytek dne,
      udržujte vzdálenost mezi vozidly, sledujte zvláštní ukazatele pro
      objížďku, volný text</MTXT>
      <MEVT>
        <TMCE directionalityvalue="1" diversion="true" durationtext="po zbytek dne"␣
↪timescalevalue="D" urgencyvalue="U">
          <EVI eventcode="980" eventorder="1" updateclass="5">
            <TXUCL language="CZ">Dopravní uzavírky a omezení</TXUCL>
            <TXEVC language="CZ">neprůjezdné, překážka na vozovce</TXEVC>
          </EVI>
          <EVI eventcode="102" eventorder="2" updateclass="1">
            <TXUCL language="CZ">Dopravní situace</TXUCL>
            <TXEVC language="CZ">dopravní kolaps v úseku 1 km</TXEVC>
```

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```
        </EVI>
        <EVI eventcode="1685" eventorder="3" updateclass="38">
          <TXUCL language="CZ">Předpověď zdržení</TXUCL>
          <TXEVC language="CZ">mimořádná událost, očekávejte
          zdržení</TXEVC>
        </EVI>
        <SPI supinfocode="13" supinfotext="udržujte vzdálenost mezi vozidly"/>
        <DIV diversioncode="61" diversiontext="sledujte zvláštní ukazatele pro␣
↪objížďku" language="CZ"/>
        <TXTMCE language="CZ">neprůjezdné, překážka na vozovce, dopravní
        kolaps v úseku 1 km, mimořádná událost, očekávejte zdržení, po
        zbytek dne, udržujte vzdálenost mezi vozidly, sledujte zvláštní
        ukazatele pro objížďku</TXTMCE>
      </TMCE>
      <OTXT>volný text</OTXT>
    </MEVT>
    <MLOC>
      <TXPL>Z ulice Vídeňská - Merhautova, do ulice Provazníkova</TXPL>
      <SNTL coordsystem="S-JTSK" count="5">
        <COORD x="-599220" y="-1163113"/>
        <STEL el_code="725704"/>
        <STEL el_code="725706"/>
        <STEL el_code="638420"/>
        <STEL el_code="638412"/>
        <STEL el_code="638377"/>
      </SNTL>
    </MLOC>
    <MDST>
      <DEST CountryName="Česká republika" RegionCode="116" RegionName="Jihomoravský
↪" TownCode="582786" TownDistrictCode="550973" TownDistrictName="Brno-střed"␣
↪TownName="Brno" TownShip="Brnoměsto" TownShipCode="3702">
        <STRE StreetCode="22063" StreetName="Cejl"/>
        <STRE StreetCode="23035" StreetName="Dvorského"/>
        <STRE StreetCode="28339" StreetName="Malinovského náměstí"/>
        <STRE StreetCode="28916" StreetName="Nové sady"/>
        <STRE StreetCode="30007" StreetName="Polní"/>
        <STRE StreetCode="30031" StreetName="Ponávka"/>
        <STRE StreetCode="30481" StreetName="Příkop"/>
        <STRE StreetCode="32131" StreetName="Renneská třída"/>
        <STRE StreetCode="33081" StreetName="Nádražní"/>
        <STRE StreetCode="33529" StreetName="Benešova"/>
        <STRE StreetCode="33537" StreetName="Milady Horákové"/>
        <STRE StreetCode="34584" StreetName="Vídeňská"/>
        <STRE StreetCode="34932" StreetName="Vsetínská"/>
        <STRE StreetCode="36871" StreetName="Strážní"/>
      </DEST>
      <DEST CountryName="Česká republika" RegionCode="116" RegionName="Jihomoravský
↪" TownCode="582786" TownDistrictCode="551031" TownDistrictName="Brno-sever"␣
↪TownName="Brno" TownShip="Brnoměsto" TownShipCode="3702">
        <STRE StreetCode="27791" StreetName="Merhautova"/>
        <STRE StreetCode="33537" StreetName="Milady Horákové"/>
      </DEST>
    </MDST>
    <DIVLOC>
      <DIVROUTE description="pro osobní automobily">
        <TXPL>textový popis trasy objížďky</TXPL>
```

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```
        </DIVROUTE>
      </DIVLOC>
    </MSG>
  </MJD>
</DOC>
```

CHAPTER

FIVE

APPENDIX B: REPRESENTATIVE XML SCHEMA: RELAX NG

We use RELAX NG Compact syntax to describe the XML document structure and we recommend you to try reading it in this form as it seems to be the most compact and readable schema format we have found.

Anyway, if you insist on using XML Schema 1.0, we include also this sort of schema produced by conversion from RELAX NG.

5.1 RELAX NG Compact Syntax Schema (RNC)

RNC schema describes expected structures of XML documents.

Root element is introduced by *start =*

All other *<SomeName> =* elements define named patterns.

In our schema there are following sections:

- prolog (namespace declaration)
- includes (not used here)
- root element definition (*start =*)
- patterns for elements
- patterns for attributes
- patterns for data types

Comments ## specify more details about meaning of particular parts.

5.1.1 *TrafficInformation.rnc* - root of the schema

```
default namespace = ""

start =

## Root element of the document
element DOC {

    ## Format version
    attribute version { type-formatVersion },

    ## Document GUID string
    ##
    ## The format of the GUID is not specified. Any format
    ## (6-4, 6-4-4-4-6, 8-4-4-4-12, number, ...) is acceptable,
    ## however it has to be unique.
```

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```
    attribute id { text },

    ## Country code
    ##
    ## One document can only contain events from one country
    attribute country { type-country },

    ## Data set type
    ##
    ## Always "extended"
    attribute DataSet { "extended" },

    ## Metadata regarding subjects involved
    ##
    ## Also contains data set versions
    element INF {

        ## ID of the organization supplying the TI
        ##
        ## Example: "JSDI_NDIC"
        attribute sender { text },

        ## ID of the organization for which the TI are provided
        attribute receiver { text },

        ## Transfer protocol
        attribute transmission { "HTTP" | "SMTP" | "FTP" | "SOAP" },

        ## Metadata regarding data sets used in the document
        element DAT {

            ## Alert-C Event List
            element EVTT {

                ## Version
                attribute version { type-AlertCEventTableVersion },

                ## Language
                attribute language { type-language-cz }
            },

            ## Alert-C Location Table
            # TODO: Missing from the format specification
            # TODO: Not present in all real-world samples
            element LOCT {

                ## Version
                attribute version { type-AlertCLocationTableVersion },

                ## Location Table Number
                attribute number { xsd:positiveInteger },

                ## County Code
                attribute country { xsd:positiveInteger }
            }?,
```

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```
            ## Street network metadata
            ##
            ## Mandatory if the document makes use of referencing to any street␣
↪network
            element SNET {

                ## Street Network Type
                ##
                ## "SN" = streetnet "GN" = GlobalNetwork
                attribute type { type-snet-type },

                ## Version
                attribute version { type-networkVersion },

                ## Country
                attribute country { "CZ" }
            }?,

            ## Metadata regarding UIRADR (Czech Street Addresses Database)
            # TODO: The format specification names this element 'UIADR' (spelling␣
↪mistake likely)
            # TODO: The format specification permits to omit this element if the␣
↪document does not
            # make use of the UIRADR database
            element UIRADR {

                ## Structure version
                attribute structure { text },

                ## Data version
                attribute version { type-nonZeroDecimal },

                ## Date of publication
                attribute date { xsd:date }?
            }
        }
    },

    ## Container for all traffic information (TI) Messages
    element MJD {

        ## Number of TI messages in this document
        attribute count { xsd:positiveInteger },

        ## Container for a single TI Message
        element-MSG*
    }
}

############################################################

element-MSG =

## Container for a single TI Message
element MSG {
```

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```
    ## An unique TI message identifier (GUID)
    ##
    ## The format of the GUID is arbitrary.
    attribute id { text },

    ## TI Message Version
    ##
    ## New = 1; any subsequential update +1
    # TODO: Messages with version = "0" do exist in the real-world samples.
    attribute version { xsd:nonNegativeInteger },

    ## Type of the TI Message
    ##
    ## "TI" - Traffic Information
    ## "TL" - Traffic Level
    # Type "WCOND" is described in a separate file
    attribute type { "TI" | "TL" },

    ## Character of the TI Message
    ##
    ## "true"  ... Planned
    ## "false" ... Unplanned
    attribute planned { xsd:boolean },

    ## Date and time attributes of the TI Message
    element MTIME {

        ## Date time format
        ##
        ## Fixed to the ISO 8601 date and time format
        attribute format { "YYYY-MM-DDThh:mm:ssTZD" },

        ## Time and date when the TI Message was generated
        element TGEN { xsd:dateTime },

        ## Time and date of the START of the event described in the TI Message
        element TSTA { xsd:dateTime },

        ## Time and date of the END of the event described in the TI Message
        element TSTO { xsd:dateTime }
    },

    ## Textual description of the TI Message
    element MTXT {

        ## Language
        attribute language { type-language-cz },

        ## The description itself
        text
    },

    ## Event-related data
    element-MEVT,

    ## Location-related data
```

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```
    element-MLOC,

    ## Location-related data (with regard to UIRADR database)
    # TODO: The format specification permits to omit this element if the document␣
↪does not
    # make use of the UIRADR database
    element-MDST?,

    ## Diversion-related data
    ##
    ## Mandatory only if there is a diversion specified in the document
    element-DIVLOC?
}

############################################################

element-MEVT =

## Container for the event-related data in the TI Message
element MEVT {

    ## Description of the event in the Alert-C format
    ##
    ## For more information refer to the ISO/DIS 14819-1:
    ## "Intelligent transport systems - Traffic and travel information
    ## messages via traffic message coding"
    ##
    ## Mandatory if the TI Message contains data encoded in the Alert-C format
    element TMCE {

        ## Urgency of the event
        ##
        ## "N" - Normal urgency
        ## "U" - Urgent
        ## "X" - extremely Urgent
        attribute urgencyvalue { "N" | "U" | "X" },

        ## Directionality of the event
        ##
        ## "1": only one direction of traffic affected
        ## "2": both directions affected by the event
        attribute directionalityvalue { "1" | "2" },

        ## Duration type of the event
        ##
        ## "D" or "(D)" - dynamic
        ## "L" or "(L)" - longer lasting
        # TODO: The format specification refers to it as a "positive integer"
        # TODO: Empty value is also used in samples
        attribute timescalevalue { type-time-scale-value },

        ## Textual description of the TI Message duration
        ##
        ## Mandatory if the duration of the TI Message is specified by a Duration Code
        attribute durationtext { text }?,

```

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```
        ## Whether a Diversion is specified
        ##
        # TODO: The format specification refers to "True" or "False" values.
        # However real-world samples use "true" or "false".
        attribute diversion { xsd:boolean },

        ## Event codes according to 14819-2: 3.1.3 Event list
        ##
        ## It can be repeated up to three times
        element EVI {

            ## Event code
            attribute eventcode { xsd:positiveInteger },

            ## Update class
            ##
            ## According to the Alert-C Event List
            attribute updateclass { xsd:positiveInteger },

            ## Quantifier
            ##
            ## Mandatory if a quantifier is used
            attribute quantifier { xsd:nonNegativeInteger }?,

            ## Event index
            attribute eventorder { xsd:integer { minInclusive="1" maxInclusive="3" } }
↪,

            ## Textual description of the Update Class of the Event
            element TXUCL {

                ## Language
                attribute language { type-language-cz },

                ## The description
                text
            },

            ## Textual description of the Event
            element TXEVC {

                ## Language
                attribute language { type-language-cz },

                ## The description
                text
            }
        }+,

        ## Supplemental information according to
        ## 14819-2: 3.2 Supplementary information
        ##
        ## Mandatory if a Supplementary Information was specified
        element SPI {

            ## Supplementary Information Code
```

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```
            ##
            ## According to 14819-2: 3.2.2 Supplementary information list
            attribute supinfocode { xsd:positiveInteger },

            ## Supplementary Information Text
            attribute supinfotext { text },

            ## Speed Limit Code
            ##
            ## Mandatory if specified
            attribute speedlimit { xsd:integer { minInclusive="1" maxInclusive="26" }␣
↪}?,

            ## Affected length accroding to
            ## Annex List of Quantifiers 14819-2: Alert-C
            ##
            ## 0 = longer than 100 km;
            ## 1-10 = between 1 to 10 km (with 1 km increment);
            ## 11-15 = between 12 to 20 km (with 2 km increment);
            ## 16-31 = between 25 to 100 km (with 5 km increment);
            ##
            ## Mandatory if specified
            attribute length { xsd:integer { minInclusive="0" maxInclusive="31" } }?
        }?,

        ## Diversion information
        ##
        ## Mandatory if Diversion was specified
        element DIV {

            ## Diversion Code
            ## according to 14819-2: 3.2.2 Supplementary information list
            ##
            ## Mandatory if Diversion specified
            attribute diversioncode { xsd:positiveInteger }?,

            ## Textual description
            ##
            ## If no Diversion Code is specified, then the defalut value is
            ##  "Diversion recomended"
            attribute diversiontext { text },

            ## Language
            attribute language { type-language-cz }
        }?,

        ## Textual description of the Alert-C Message
        element TXTMCE {

            ## Language
            attribute language { type-language-cz },

            ## Message
            text
        }
    }?,
```

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```
    ## Free text supplied by the TI Message provider
    ##
    ## Mandatory on if present in the message
    element OTXT {

        ## Message
        text
    }?
}

############################################################

element-MLOC =

## Location-related data
element MLOC {

    ## Textual description of the location or suggested diversion
    element-TXPL,

    ## Location specified using Alert-C Location Table
    element TMCL {

        ## Alert-C Primary Location
        ##
        ## The format specification refers to it as "Primarycode"
        attribute primarycode { xsd:positiveInteger },

        ## Alert-C Extent
        ## The format specification refers to it as "Extent"
        attribute extent { xsd:integer { minInclusive="0" maxInclusive="32" } },

        ## Alert-C Direction
        ## The format specification refers to it as "Direction"
        attribute direction { "+" | "-" },

        ## Road ID
        ## The format specification refers to it as "Roadid"
        attribute roadid { xsd:positiveInteger }
    }?,

    ## Location specified by a road network
    element SNTL {

        ## Coordinate system used
        attribute coordsystem { type-coord-system },

        ## Number of road elements affected
        ## The format specification refers to it as "Count"
        attribute count { xsd:positiveInteger },

                ## Coordinates of the beginning of the event in the sense of traffic
                ## direction, in the specified coordinate system. For linear␣
↪locations it is the tail of
                ## a traffic event, the place of first arrival or the secondary TMC␣
```

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```
↪location).
                ##
        element COORD {

            ## x-coord
            attribute x { xsd:decimal },

            ## x-coord
            attribute y { xsd:decimal }
        },

        ## Road network elements affected
        element STEL {

            ## ID of the road element
            attribute el_code { xsd:positiveInteger }
        }+
    }
}

############################################################

element-MDST =

## Location-related data (with regard to UIRADR database)
# TODO: Mandatory if the message uses UIRADR database
element MDST {

    ## Container for the information related to one administration unit
    ##
    ## Can be repeated if the TI Message affects more administration units
    element DEST {

        ## Country Name
        attribute CountryName { text },

        ## Part of the Town
        ##
        ## Mandatory if it exists
        attribute TownDistrictName { text }?,

        ## Code of the Part of the Town
        ##
        ## Mandatory if it exists
        attribute TownDistrictCode { xsd:integer }?,

        ## Town Name
        attribute TownName { text },

        ## Town Code
        attribute TownCode { xsd:positiveInteger },

        ## District Name
        attribute TownShip { text },

        ## District Code
```

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```
        attribute TownShipCode { xsd:positiveInteger },

        ## Region Name
        attribute RegionName { text },

        ## Region Code
        attribute RegionCode { xsd:positiveInteger },

        ## Location witch regard to Streets
        ##
        ## Mandatory for line-type event
        ##
        ## Repeating names is not allowed
        ##
        ## One STRE element will be generated for each Street affected by the event
        element STRE {

            ## Street Name
            ##
            ## Manadatory if available
            attribute StreetName { text }?,

            ## Street Code
            ##
            ## Mandatory if available
            attribute StreetCode { xsd:positiveInteger }?
        }*,

        ## Location witch regard to UIRADR
        ##
        ## Mandatory for line-type event
        ##
        ## Repeating names is not allowed
        ##
        ## One ROAD element will be generated for each road affected by the event
        element ROAD {

            ## Road Number
            ##
            ## Mandatory if available
            attribute RoadNumber { text }?,

            ## Road Class
            ##
            ## This classification is Czech specific
            attribute RoadClass {

                ## Motorway
                "0" |

                ## 1st Class Road
                "1" |

                ## 2nd Class Road
                "2" |
```

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```
                ## 3rd Class Road
                "3" |

                ## Other roads
                "4" |

                ## Semi-motorway
                "5"
            }
        }*
    }+
}

############################################################

element-DIVLOC =

## Diversion
##
## Mandatory only if a diversion is specified
element DIVLOC {

    ## Textual description of a leg of a diversion
    element DIVROUTE {

        ## Description
        attribute description { text }
    }+
}

############################################################

element-TXPL =

    ## Textual description of the location or suggested diversion
    element TXPL { text }

############################################################

type-nonZeroDecimal = xsd:decimal { minExclusive="0.0" }

type-formatVersion = type-nonZeroDecimal

type-version = xsd:positiveInteger

type-AlertCEventTableVersion = type-nonZeroDecimal

type-AlertCLocationTableVersion = type-nonZeroDecimal

type-language-cz = "CZ"

type-country = "CZ" | "AT" | "DE" | "SK" | "PL"

type-snet-type = "SN" | "GN"

type-networkVersion = type-nonZeroDecimal
```

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```
type-time-scale-value = "D" | "(D)" | "L" | "(L)" | ""

type-coord-system = "S-JTSK" | "WGS-84"
```

CHAPTER

SIX

CHANGELOG

version 3.2.5.0

- Initial format and documentation

CHAPTER

SEVEN

APPENDIX C: ALTERNATIVE XML SCHEMA: W3C XML SCHEMA 1.0 (XSD)

W3C XML Schema 1.0 is probably the most popular format for describing XML structures today.

XSD schema provided bellow was created by transforming RNC schema and shall reflect all the features from RNC. In (unlikely) case, there are differences, RELAX NG Compact syntax is the one to use.

7.1 *TrafficInformation.xsd* - root of the schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DOC">
    <xs:annotation>
      <xs:documentation>Root element of the document</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="INF"/>
        <xs:element ref="MJD"/>
      </xs:sequence>
      <xs:attribute name="version" type="type-formatVersion" use="required">
        <xs:annotation>
          <xs:documentation>Format version</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="id" use="required">
        <xs:annotation>
          <xs:documentation>Document GUID string The format of the GUID is
          not specified. Any format (6-4, 6-4-4-4-6, 8-4-4-4-12, number,
          ...) is acceptable, however it has to be
          unique.</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="country" type="type-country" use="required">
        <xs:annotation>
          <xs:documentation>Country code One document can only contain
          events from one country</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="DataSet" use="required">
        <xs:annotation>
          <xs:documentation>Data set type Always
          &quot;extended&quot;</xs:documentation>
        </xs:annotation>
```

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```
        <xs:simpleType>
          <xs:restriction base="xs:token">
            <xs:enumeration value="extended"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="INF">
    <xs:annotation>
      <xs:documentation>Metadata regarding subjects involved. Also contains
      data set versions</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="DAT"/>
      </xs:sequence>
      <xs:attribute name="sender" use="required">
        <xs:annotation>
          <xs:documentation>ID of the organization supplying the TI
          Example: &quot;JSDI_NDIC&quot;</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="receiver" use="required">
        <xs:annotation>
          <xs:documentation>ID of the organization for which the TI are
          provided</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="transmission" use="required">
        <xs:annotation>
          <xs:documentation>Transfer protocol</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:token">
            <xs:enumeration value="HTTP"/>
            <xs:enumeration value="SMTP"/>
            <xs:enumeration value="FTP"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="DAT">
    <xs:annotation>
      <xs:documentation>Metadata regarding data sets used in the
      document</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="EVTT"/>
        <xs:element minOccurs="0" ref="LOCT"/>
        <xs:element minOccurs="0" ref="SNET"/>
        <xs:element ref="UIRADR"/>
      </xs:sequence>
    </xs:complexType>
```

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```
</xs:element>
<xs:element name="EVTT">
  <xs:annotation>
    <xs:documentation>Alert-C Event List</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute name="version" type="type-AlertCEventTableVersion" use="required">
      <xs:annotation>
        <xs:documentation>Version</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="language" type="type-language-cz" use="required">
      <xs:annotation>
        <xs:documentation>Language</xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="LOCT">
  <xs:annotation>
    <xs:documentation>Alert-C Location Table</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute name="version" type="type-AlertCLocationTableVersion" use=
↪"required">
      <xs:annotation>
        <xs:documentation>Version</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="number" type="xs:positiveInteger" use="required">
      <xs:annotation>
        <xs:documentation>Location Table Number</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="country" type="xs:positiveInteger" use="required">
      <xs:annotation>
        <xs:documentation>County Code</xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="SNET">
  <xs:annotation>
    <xs:documentation>Street network metadata Mandatory if the document
    makes use of referencing to any street network</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute name="type" type="type-snet-type" use="required">
      <xs:annotation>
        <xs:documentation>Street Network Type &quot;SN&quot; = streetnet &quot;GN&
↪quot; =
        GlobalNetwork</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="version" type="type-networkVersion" use="required">
      <xs:annotation>
```

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```
          <xs:documentation>Version</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="country" use="required">
        <xs:annotation>
          <xs:documentation>Country</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:token">
            <xs:enumeration value="CZ"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="UIRADR">
    <xs:annotation>
      <xs:documentation>Metadata regarding UIRADR (Czech Street Addresses
      Database)</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attribute name="structure" use="required">
        <xs:annotation>
          <xs:documentation>Structure version</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="version" type="type-nonZeroDecimal" use="required">
        <xs:annotation>
          <xs:documentation>Data version</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="date" type="xs:date">
        <xs:annotation>
          <xs:documentation>Date of publication</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="MJD">
    <xs:annotation>
      <xs:documentation>Container for all traffic information (TI)
      Messages</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element maxOccurs="unbounded" minOccurs="0" ref="MSG">
          <xs:annotation>
            <xs:documentation>Container for a single TI
            Message</xs:documentation>
          </xs:annotation>
        </xs:element>
      </xs:sequence>
      <xs:attribute name="count" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>Number of TI messages in this
          document</xs:documentation>
```

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```
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="MSG">
  <xs:annotation>
    <xs:documentation>Container for a single TI
    Message</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="MTIME"/>
      <xs:element ref="MTXT"/>
      <xs:element ref="MEVT">
        <xs:annotation>
          <xs:documentation>Event-related data</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element ref="MLOC">
        <xs:annotation>
          <xs:documentation>Location-related data</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element minOccurs="0" ref="MDST">
        <xs:annotation>
          <xs:documentation>Location-related data (with regard to UIRADR
          database)</xs:documentation>
        </xs:annotation>
      </xs:element>
      <xs:element minOccurs="0" ref="DIVLOC">
        <xs:annotation>
          <xs:documentation>Diversion-related data Mandatory only if
          there is a diversion specified in the
          document</xs:documentation>
        </xs:annotation>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="id" use="required">
      <xs:annotation>
        <xs:documentation>An unique TI message identifier (GUID) The
        format of the GUID is arbitrary.</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="version" type="xs:nonNegativeInteger" use="required">
      <xs:annotation>
        <xs:documentation>TI Message Version New = 1; any subsequential
        update +1</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="type" use="required">
      <xs:annotation>
        <xs:documentation>Type of the TI Message &quot;TI&quot; - Traffic
        Information &quot;TL&quot; - Traffic Level</xs:documentation>
      </xs:annotation>
      <xs:simpleType>
        <xs:restriction base="xs:token">
```

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```
          <xs:enumeration value="TI"/>
          <xs:enumeration value="TL"/>
        </xs:restriction>
      </xs:simpleType>
    </xs:attribute>
    <xs:attribute name="planned" type="xs:boolean" use="required">
      <xs:annotation>
        <xs:documentation>Character of the TI Message &quot;true&quot; ... Planned
        &quot;false&quot; ... Unplanned</xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="MTIME">
  <xs:annotation>
    <xs:documentation>Date and time attributes of the TI
    Message</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element ref="TGEN"/>
      <xs:element ref="TSTA"/>
      <xs:element ref="TSTO"/>
    </xs:sequence>
    <xs:attribute name="format" use="required">
      <xs:annotation>
        <xs:documentation>Date time format Fixed to the ISO 8601 date and
        time format</xs:documentation>
      </xs:annotation>
      <xs:simpleType>
        <xs:restriction base="xs:token">
          <xs:enumeration value="YYYY-MM-DDThh:mm:ssTZD"/>
        </xs:restriction>
      </xs:simpleType>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="TGEN" type="xs:dateTime">
  <xs:annotation>
    <xs:documentation>Time and date when the TI Message was
    generated</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="TSTA" type="xs:dateTime">
  <xs:annotation>
    <xs:documentation>Time and date of the START of the event described
    in the TI Message</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="TSTO" type="xs:dateTime">
  <xs:annotation>
    <xs:documentation>Time and date of the END of the event described in
    the TI Message</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="MTXT">
```

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```
    <xs:annotation>
      <xs:documentation>Textual description of the TI
      Message</xs:documentation>
    </xs:annotation>
    <xs:complexType mixed="true">
      <xs:attribute name="language" type="type-language-cz" use="required">
        <xs:annotation>
          <xs:documentation>Language</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="MEVT">
    <xs:annotation>
      <xs:documentation>Container for the event-related data in the TI
      Message</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element minOccurs="0" ref="TMCE"/>
        <xs:element minOccurs="0" ref="OTXT"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
  <xs:element name="TMCE">
    <xs:annotation>
      <xs:documentation>Description of the event in the Alert-C format For
      more information refer to the ISO/DIS 14819-1: &quot;Intelligent transport
      systems - Traffic and travel information messages via traffic message
      coding&quot; Mandatory if the TI Message contains data encoded in the
      Alert-C format</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element maxOccurs="unbounded" ref="EVI"/>
        <xs:element minOccurs="0" ref="SPI"/>
        <xs:element minOccurs="0" ref="DIV"/>
        <xs:element ref="TXTMCE"/>
      </xs:sequence>
      <xs:attribute name="urgencyvalue" use="required">
        <xs:annotation>
          <xs:documentation>Urgency of the event &quot;N&quot; - Normal urgency &quot;
↪U&quot; -
          Urgent &quot;X&quot; - extremely Urgent</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:token">
            <xs:enumeration value="N"/>
            <xs:enumeration value="U"/>
            <xs:enumeration value="X"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
      <xs:attribute name="directionalityvalue" use="required">
        <xs:annotation>
          <xs:documentation>Directionality of the event &quot;1&quot;: only one
```

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```
          direction of traffic affected &quot;2&quot;: both directions affected by
          the event</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:token">
            <xs:enumeration value="1"/>
            <xs:enumeration value="2"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
      <xs:attribute name="timescalevalue" type="type-time-scale-value" use="required">
        <xs:annotation>
          <xs:documentation>Duration type of the event &quot;D&quot; or &quot;(D)&
↪quot; -
          dynamic &quot;L&quot; or &quot;(L)&quot; - longer lasting</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="durationtext">
        <xs:annotation>
          <xs:documentation>Textual description of the TI Message duration
          Mandatory if the duration of the TI Message is specified by a
          Duration Code</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="diversion" type="xs:boolean" use="required">
        <xs:annotation>
          <xs:documentation>Whether a Diversion is
          specified</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="EVI">
    <xs:annotation>
      <xs:documentation>Event codes according to 14819-2: 3.1.3 Event list
      It can be repeated up to three times</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="TXUCL"/>
        <xs:element ref="TXEVC"/>
      </xs:sequence>
      <xs:attribute name="eventcode" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>Event code</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="updateclass" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>Update class According to the Alert-C Event
          List</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="quantifier" type="xs:nonNegativeInteger">
        <xs:annotation>
          <xs:documentation>Quantifier Mandatory if a quantifier is
```

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```
          used</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="eventorder" use="required">
        <xs:annotation>
          <xs:documentation>Event index</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:integer">
            <xs:minInclusive value="1"/>
            <xs:maxInclusive value="3"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="TXUCL">
    <xs:annotation>
      <xs:documentation>Textual description of the Update Class of the
      Event</xs:documentation>
    </xs:annotation>
    <xs:complexType mixed="true">
      <xs:attribute name="language" type="type-language-cz" use="required">
        <xs:annotation>
          <xs:documentation>Language</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="TXEVC">
    <xs:annotation>
      <xs:documentation>Textual description of the Event</xs:documentation>
    </xs:annotation>
    <xs:complexType mixed="true">
      <xs:attribute name="language" type="type-language-cz" use="required">
        <xs:annotation>
          <xs:documentation>Language</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="SPI">
    <xs:annotation>
      <xs:documentation>Supplemental information according to 14819-2: 3.2
      Supplementary information Mandatory if a Supplementary Information
      was specified</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attribute name="supinfocode" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>Supplementary Information Code According to
          14819-2: 3.2.2 Supplementary information list</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="supinfotext" use="required">
        <xs:annotation>
```

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```
          <xs:documentation>Supplementary Information
          Text</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="speedlimit">
        <xs:annotation>
          <xs:documentation>Speed Limit Code Mandatory if
          specified</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:integer">
            <xs:minInclusive value="1"/>
            <xs:maxInclusive value="26"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
      <xs:attribute name="length">
        <xs:annotation>
          <xs:documentation>Affected length accroding to Annex List of
          Quantifiers 14819-2: Alert-C 0 = longer than 100 km; 1-10 =
          between 1 to 10 km (with 1 km increment); 11-15 = between 12 to
          20 km (with 2 km increment); 16-31 = between 25 to 100 km (with 5
          km increment); Mandatory if specified</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:integer">
            <xs:minInclusive value="0"/>
            <xs:maxInclusive value="31"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="DIV">
    <xs:annotation>
      <xs:documentation>Diversion information Mandatory if Diversion was
      specified</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attribute name="diversioncode" type="xs:positiveInteger">
        <xs:annotation>
          <xs:documentation>Diversion Code according to 14819-2: 3.2.2
          Supplementary information list Mandatory if Diversion
          specified</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="diversiontext" use="required">
        <xs:annotation>
          <xs:documentation>Textual description If no Diversion Code is
          specified, then the defalut value is &quot;Diversion
          recomended&quot;</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="language" type="type-language-cz" use="required">
        <xs:annotation>
          <xs:documentation>Language</xs:documentation>
```

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```
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="TXTMCE">
  <xs:annotation>
    <xs:documentation>Textual description of the Alert-C
    Message</xs:documentation>
  </xs:annotation>
  <xs:complexType mixed="true">
    <xs:attribute name="language" type="type-language-cz" use="required">
      <xs:annotation>
        <xs:documentation>Language</xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="OTXT" type="xs:string">
  <xs:annotation>
    <xs:documentation>Free text supplied by the TI Message provider
    Mandatory on if present in the message</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:element name="MLOC">
  <xs:annotation>
    <xs:documentation>Location-related data</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="element-TXPL">
        <xs:sequence>
          <xs:element minOccurs="0" ref="TMCL"/>
          <xs:element ref="SNTL"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
<xs:element name="TMCL">
  <xs:annotation>
    <xs:documentation>Location specified using Alert-C Location
    Table</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute name="primarycode" type="xs:positiveInteger" use="required">
      <xs:annotation>
        <xs:documentation>Alert-C Primary Location The format
        specification refers to it as &quot;Primarycode&quot;</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="extent" use="required">
      <xs:annotation>
        <xs:documentation>Alert-C Extent The format specification refers
        to it as &quot;Extent&quot;</xs:documentation>
      </xs:annotation>
      <xs:simpleType>
```

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```
          <xs:restriction base="xs:integer">
            <xs:minInclusive value="0"/>
            <xs:maxInclusive value="32"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
      <xs:attribute name="direction" use="required">
        <xs:annotation>
          <xs:documentation>Alert-C Direction The format specification
          refers to it as &quot;Direction&quot;</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:token">
            <xs:enumeration value="+"/>
            <xs:enumeration value="-"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
      <xs:attribute name="roadid" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>Road ID The format specification refers to it
          as &quot;Roadid&quot;</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="SNTL">
    <xs:annotation>
      <xs:documentation>Location specified by a road
      network</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element ref="COORD"/>
        <xs:element maxOccurs="unbounded" ref="STEL"/>
      </xs:sequence>
      <xs:attribute name="coordsystem" type="type-coord-system" use="required">
        <xs:annotation>
          <xs:documentation>Coordinate system used</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="count" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>Number of road elements affected The format
          specification refers to it as &quot;Count&quot;</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="COORD">
    <xs:annotation>
      <xs:documentation>Coordinates of the beginning of the event in the sense of
                traffic direction, in the specified coordinate system. For linear␣
↪locations
                it is the tail of a traffic event, the place of first arrival or␣
↪the secondary TMC
```

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```
                  location).</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attribute name="x" type="xs:decimal" use="required">
        <xs:annotation>
          <xs:documentation>x-coord</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="y" type="xs:decimal" use="required">
        <xs:annotation>
          <xs:documentation>y-coord</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="STEL">
    <xs:annotation>
      <xs:documentation>Road network elements affected</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attribute name="el_code" type="xs:positiveInteger" use="required">
        <xs:annotation>
          <xs:documentation>ID of the road element</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="MDST">
    <xs:annotation>
      <xs:documentation>Location-related data (with regard to UIRADR
      database)</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element maxOccurs="unbounded" ref="DEST"/>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
  <xs:element name="DEST">
    <xs:annotation>
      <xs:documentation>Container for the information related to one
      administration unit Can be repeated if the TI Message affects more
      administration units</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:sequence>
        <xs:element maxOccurs="unbounded" minOccurs="0" ref="STRE"/>
        <xs:element maxOccurs="unbounded" minOccurs="0" ref="ROAD"/>
      </xs:sequence>
      <xs:attribute name="CountryName" use="required">
        <xs:annotation>
          <xs:documentation>Country Name</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="TownDistrictName">
        <xs:annotation>
```

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```
        <xs:documentation>Part of the Town Mandatory if it
        exists</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="TownDistrictCode" type="xs:integer">
      <xs:annotation>
        <xs:documentation>Code of the Part of the Town Mandatory if it
        exists</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="TownName" use="required">
      <xs:annotation>
        <xs:documentation>Town Name</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="TownCode" type="xs:positiveInteger" use="required">
      <xs:annotation>
        <xs:documentation>Town Code</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="TownShip" use="required">
      <xs:annotation>
        <xs:documentation>District Name</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="TownShipCode" type="xs:positiveInteger" use="required">
      <xs:annotation>
        <xs:documentation>District Code</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="RegionName" use="required">
      <xs:annotation>
        <xs:documentation>Region Name</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="RegionCode" type="xs:positiveInteger" use="required">
      <xs:annotation>
        <xs:documentation>Region Code</xs:documentation>
      </xs:annotation>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="STRE">
  <xs:annotation>
    <xs:documentation>Location witch regard to Streets Mandatory for
    line-type event Repeating names is not allowed One STRE element will
    be generated for each Street affected by the event</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:attribute name="StreetName">
      <xs:annotation>
        <xs:documentation>Street Name Manadatory if
        available</xs:documentation>
      </xs:annotation>
    </xs:attribute>
    <xs:attribute name="StreetCode" type="xs:positiveInteger">
```

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```
        <xs:annotation>
          <xs:documentation>Street Code Mandatory if
          available</xs:documentation>
        </xs:annotation>
      </xs:attribute>
    </xs:complexType>
  </xs:element>
  <xs:element name="ROAD">
    <xs:annotation>
      <xs:documentation>Location witch regard to UIRADR Mandatory for
      line-type event Repeating names is not allowed One ROAD element will
      be generated for each road affected by the event</xs:documentation>
    </xs:annotation>
    <xs:complexType>
      <xs:attribute name="RoadNumber">
        <xs:annotation>
          <xs:documentation>Road Number Mandatory if
          available</xs:documentation>
        </xs:annotation>
      </xs:attribute>
      <xs:attribute name="RoadClass" use="required">
        <xs:annotation>
          <xs:documentation>Road Class This classification is Czech
          specific</xs:documentation>
        </xs:annotation>
        <xs:simpleType>
          <xs:restriction base="xs:token">
            <xs:enumeration value="0">
              <xs:annotation>
                <xs:documentation>Motorway</xs:documentation>
              </xs:annotation>
            </xs:enumeration>
            <xs:enumeration value="1">
              <xs:annotation>
                <xs:documentation>1st Class Road</xs:documentation>
              </xs:annotation>
            </xs:enumeration>
            <xs:enumeration value="2">
              <xs:annotation>
                <xs:documentation>2nd Class Road</xs:documentation>
              </xs:annotation>
            </xs:enumeration>
            <xs:enumeration value="3">
              <xs:annotation>
                <xs:documentation>3rd Class Road</xs:documentation>
              </xs:annotation>
            </xs:enumeration>
            <xs:enumeration value="4">
              <xs:annotation>
                <xs:documentation>Other roads</xs:documentation>
              </xs:annotation>
            </xs:enumeration>
            <xs:enumeration value="5">
              <xs:annotation>
                <xs:documentation>Semi-motorway</xs:documentation>
              </xs:annotation>
```

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```xml
          </xs:enumeration>
        </xs:restriction>
      </xs:simpleType>
    </xs:attribute>
  </xs:complexType>
</xs:element>
<xs:element name="DIVLOC">
  <xs:annotation>
    <xs:documentation>Diversion Mandatory only if a diversion is
    specified</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:sequence>
      <xs:element maxOccurs="unbounded" ref="DIVROUTE"/>
    </xs:sequence>
  </xs:complexType>
</xs:element>
<xs:element name="DIVROUTE">
  <xs:annotation>
    <xs:documentation>Textual description of a leg of a
    diversion</xs:documentation>
  </xs:annotation>
  <xs:complexType>
    <xs:complexContent>
      <xs:extension base="element-TXPL">
        <xs:attribute name="description" use="required">
          <xs:annotation>
            <xs:documentation>Description</xs:documentation>
          </xs:annotation>
        </xs:attribute>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
<xs:complexType name="element-TXPL">
  <xs:annotation>
    <xs:documentation/>
  </xs:annotation>
  <xs:sequence>
    <xs:element ref="TXPL"/>
  </xs:sequence>
</xs:complexType>
<xs:element name="TXPL" type="xs:string">
  <xs:annotation>
    <xs:documentation>Textual description of the location or suggested
    diversion</xs:documentation>
  </xs:annotation>
</xs:element>
<xs:simpleType name="type-nonZeroDecimal">
  <xs:annotation>
    <xs:documentation/>
  </xs:annotation>
  <xs:restriction base="xs:decimal">
    <xs:minExclusive value="0.0"/>
  </xs:restriction>
</xs:simpleType>
```


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```
  <xs:simpleType name="type-formatVersion">
    <xs:restriction base="type-nonZeroDecimal"/>
  </xs:simpleType>
  <xs:simpleType name="type-version">
    <xs:restriction base="xs:positiveInteger"/>
  </xs:simpleType>
  <xs:simpleType name="type-AlertCEventTableVersion">
    <xs:restriction base="type-nonZeroDecimal"/>
  </xs:simpleType>
  <xs:simpleType name="type-AlertCLocationTableVersion">
    <xs:restriction base="type-nonZeroDecimal"/>
  </xs:simpleType>
  <xs:simpleType name="type-language-cz">
    <xs:restriction base="xs:token">
      <xs:enumeration value="CZ"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="type-country">
    <xs:restriction base="xs:token">
      <xs:enumeration value="CZ"/>
      <xs:enumeration value="AT"/>
      <xs:enumeration value="DE"/>
      <xs:enumeration value="SK"/>
      <xs:enumeration value="PL"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="type-snet-type">
    <xs:restriction base="xs:token">
      <xs:enumeration value="SN"/>
      <xs:enumeration value="GN"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="type-networkVersion">
    <xs:restriction base="type-nonZeroDecimal"/>
  </xs:simpleType>
  <xs:simpleType name="type-time-scale-value">
    <xs:restriction base="xs:token">
      <xs:enumeration value="D"/>
      <xs:enumeration value="(D)"/>
      <xs:enumeration value="L"/>
      <xs:enumeration value="(L)"/>
      <xs:enumeration value=""/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="type-coord-system">
    <xs:restriction base="xs:token">
      <xs:enumeration value="S-JTSK"/>
      <xs:enumeration value="WGS-84"/>
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
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